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RESEARCH ARTICLE

#GAMEADDICTED: A Machine Learning Framework for Digital Game Addiction Detection and Early Intervention

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ABSTRACT This study addresses the growing concern of digital game addiction (DGA) by developing an information system for detection and early intervention, to predict an individual’s propensity toward DGA. The study employed machine learning models to process unstructured data collected from online game platforms between May 1, 2023, and June 1, 2023. A total of 660 comments were automatically extracted. Following an expert review process, 245 ambiguous or inconsistent entries were removed, resulting in a final dataset of 415 textual comments for model training. The comments were manually annotated into two categories: addiction-related (1) and non-addiction-related (0) to ensure data consistency and reliability. Data augmentation techniques, such as BERT-based augmentation, deep translation, and synonym replacement, were applied to increase dataset diversity and enhance the model’s generalization capability. After extensive preprocessing, we trained and evaluated six machine learning models: XGBoost, Support Vector Machines (SVM), Logistic Regression, K-Nearest Neighbors (KNN), Decision Trees, and Random Forests. XGBoost consistently outperformed all other models, achieving the highest accuracy of 93.06% on the binary-labeled dataset and 92.49% on the expanded dataset. Random Forests and Logistic Regression also exhibited strong classification performance, making them viable alternatives depending on the requirements. In contrast, SVM and Decision Trees showed lower predictive reliability, indicating potential challenges in classifying text-based addiction indicators effectively. As a practical outcome, we developed an interface using the best-performing model, capable of performing preliminary diagnoses based on user-provided sentences. This innovative approach introduces a new dimension in addiction studies by presenting a practical tool for early detection and intervention in DGA.

INDEX TERMS Digital game addiction, data analytics, early diagnosis, machine learning, text classification, natural language processing, human computer interaction, social informatics.

I. INTRODUCTION

A. MOTIVATION

Digital games have become a prevalent form of entertainment, stress relief, and social interaction in modern society. However, excessive and unregulated gaming can lead to Digital Game Addiction (DGA), a behavioral disorder

associated with significant mental, physical, and social consequences [1]. Similar to other forms of behavioral addiction, DGA can result in social isolation, sleep disturbances, depression, and cognitive impairments. If left unmanaged, individuals with DGA may develop compulsive gaming behaviors, neglecting personal, academic, and professional responsibilities. As DGA continues to rise as a public health concern, there is a growing need for scalable and automated detection mechanisms to facilitate timely intervention.

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Traditional assessment methods, such as self-reported surveys and clinical evaluations, are often subjective and time-intensive, limiting their effectiveness for large-scale identification. This study addresses this challenge by employing machine learning techniques to automate the early detection of DGA. By developing data-driven models, this research aims to enhance scalable intervention strategies and provide valuable insights for researchers, mental health professionals, and policymakers seeking to mitigate the negative impacts of DGA.

While existing methods such as psychological counseling, cognitive-behavioral therapy, and survey-based assessments provide valuable insights [2], they often suffer from limitations, including reliance on self-reported data, high costs, and accessibility constraints. These traditional approaches can be subjective, time-consuming, and impractical for large-scale early intervention. To address these challenges, this study leverages machine learning models to analyze textual data objectively, offering a scalable and data-driven approach to DGA detection. Machine learning has been increasingly employed in various fields to analyze large volumes of unstructured text data, uncovering patterns that may not be immediately evident through conventional analytical methods. The rapid expansion of online gaming communities has resulted in a wealth of user-generated textual content, providing a unique opportunity to analyze addiction-related behaviors through natural language processing (NLP). Unlike traditional assessment methods, machine learning models can systematically extract insights from these texts, enabling early detection and intervention strategies. Machine learning models [3] further enhance these analyses, providing valuable tools to identify individuals predisposed to DGA. Therefore, the collection, preprocessing, and modeling of unstructured data form the foundation of a novel approach to combating DGA. This research aims to explore online feedback from digital game addicts.

B. CONTRIBUTIONS

This paper proposes a novel machine learning-based framework to detect DGA through short text inputs. This study aims to develop and evaluate a machine learning-based information system that identifies DGA tendencies using NLP techniques applied to user-generated textual data. A comparative analysis of widely used classification models—XGBoost, SVM, Logistic Regression, KNN, Decision Trees, and Random Forests—was conducted to determine the most effective model. The highest-performing model was then integrated into an interactive user interface, allowing individuals to self-assess their risk of DGA based on textual input.

Figure 1 presents the research process, beginning with data collection from various sources, followed by preprocessing steps such as NLP, an artificial intelligence field related to computer understanding, processing, and generation of human language. The preprocessed dataset is then used to train and evaluate multiple machine learning models, with

the best-performing model integrated into a user interface to detection of DGA. The research followed a structured approach: initially, comments from online platforms related to DGA and those advocating against it were collected and categorized as 1 and 0, respectively. Subsequently, this dataset was used to analyze widely recognized classification algorithms, including xgboost, support vector machines (SVMs), logistic regression, K-Nearest Neighbors (KNN), Decision Trees, and Random Forests. The performances were assessed through various metrics such as accuracy score, precision score, recall score, F1 score, and confusion matrix. By utilizing machine learning models, the study presents a model to predict the propensity for DGA in individuals suspected of it, based on short text inputs. As a result, a webpage was designed using the developed model, enabling preliminary diagnoses by analyzing input sentences.

Although the machine learning models employed are widely recognized, the novelty of this study lies in their application to short, real-world user comments for the detection of DGA, and in the development of an interactive interface that supports early awareness and intervention.

C. ORGANIZATION OF THE PAPER

The paper is structured as follows: The Research Background section provides a comprehensive overview of the existing literature. The Method Section details the methodologies employed in the study. The Data Collection part describes the process and sources from which the data were gathered, including data augmentation and sentiment analysis to enhance the dataset. The Data Preprocessing section discusses the techniques used to prepare the data for modeling. The Classification and Modeling section elucidates the methods for classifying and modeling the data. The Results section presents the findings of the research. Finally, the Conclusions section summarizes the key insights and implications derived from the study.

II. RESEARCH BACKGROUND

In the last 15 years, the concept of digital addiction (DA) has gained increasing recognition and attention as technology has advanced. DA refers to the problematic use of technology, characterized by excessive, impulsive, obsessive, and compulsive behaviors. DGA has emerged as a significant concern within this realm, particularly among young individuals. DGA is classified as a “Gaming Disorder” in the World Health Organization’s International Classification of Diseases and is categorized as an “Internet Gaming Disorder” in the Diagnostic and Statistical Manual of Mental Disorders [4]. The misuse of DGA can pose a significant threat, especially to young individuals.

DGA has been evolving with the advancement of technology and has held a significant place in the literature across different disciplines since the 2000s. Various studies have been conducted during this period, yielding important findings on DGA. The literature has developed in parallel

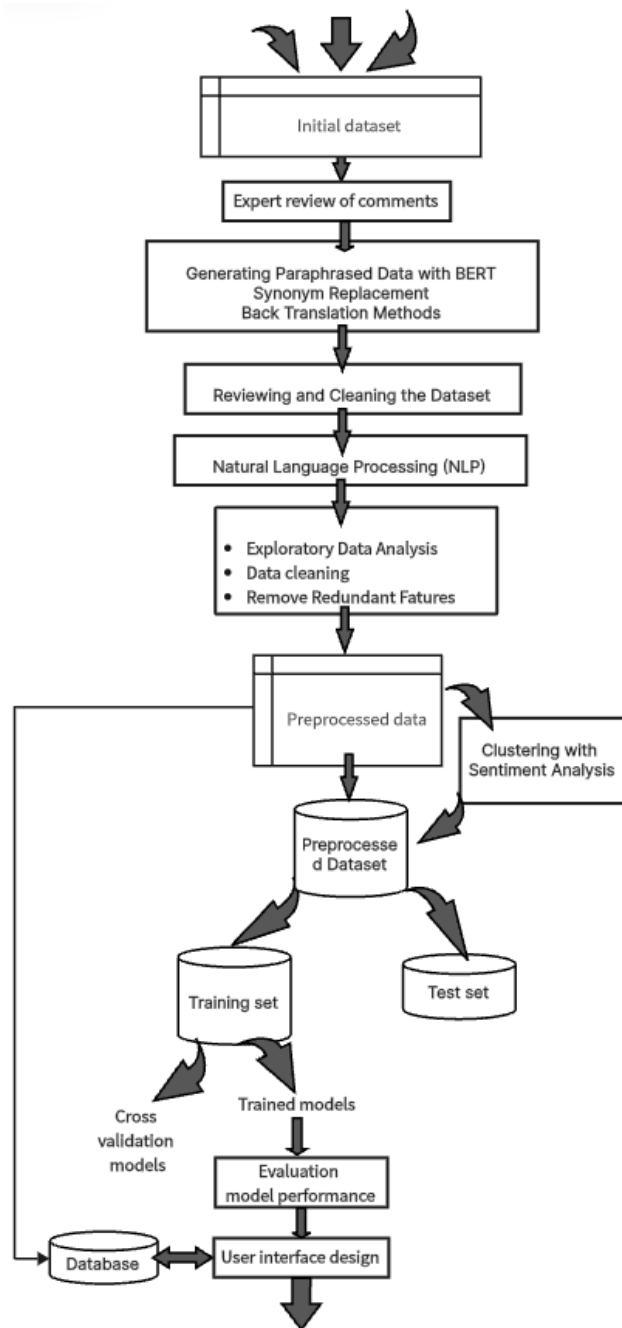


FIGURE 1. The research process.

with technological advancements, with numerous studies conducted in psychology, economics, and sociology.

Various studies examine the psychological, social, and economic dimensions of DGA, highlighting its negative impacts on individuals and proposing solutions. Chibbaro et al. [5] emphasized collaborative efforts between school personnel and counselors in integrating technology and curbing DA. Since the 2020s, understanding and addressing the phenomenon of DGA has gained traction among scholars.

Abendroth et al. [6] scrutinized various scales for measuring DGA and found significant similarities, highlighting the need for these scales to be improved with digital technology. Yengin [7] drew attention to the dangers of DGA through a survey aimed at primary school students in Istanbul. Annarumma et al. [8] identified a prevalent use of mobile devices among families, suggesting it escalates addiction levels in children and might adversely impact early childhood development. Parent-child dynamics were examined by Keya et al. [9], who noted factors like parental neglect and children's feelings of loneliness as potential catalysts for DGA.

Lazarinis et al. [10] introduced an application that employs storytelling to bolster students' awareness of online risks. Exploring more peripheral aspects of online habits, Bulut and Tuncay [11] drew connections between social media addiction and sleep issues among dental students, surprisingly finding female students to be more addicted. Gabriel et al. [12] assessed parents' attitudes toward DGA, revealing a consensus of allowing children access to digital games at schools but under strict supervision. Carter et al. [13] delved into participants' perspectives on DGA and its interaction with their identities. In the literature, DGA was addressed as a growing issue, particularly among young individuals, and its effects have become more pronounced with technological advancements. Lee et al. [14] evaluated the potential of a VR-based program designed to identify patterns in users' leisure time related to digital games. In a cultural comparison, Kaya Erdem and Civan Kemiksiz [15] examined why Generations Y and Z are dubbed the "digital gamer generation" and how this influences their career choices. Kloker et al. [16] analyzed factors affecting mobile technology adoption, pointing out the inhibiting role of privacy concerns and the escalating role of smartphone addiction. Tutgun-Unal [17] contrasted the social media addiction levels and expectations from social media among university students in South Korea and Turkey, with the former showing higher tendencies. Malik et al. [18] explored the role of parental attachment variables in shaping children's self-control behaviors and game addiction. Pogozhina [19] focused on the logical-categorical characteristics and motivation patterns of internet users' online antisocial behaviors, identifying both internal and external determinants and linking pathological internet use with significant psychological traits. With developments in social media, user feedback analysis has also started to be utilized in the field. Schatto-Eckrodt et al. [20] analyzed tweets before and after the World Health Organization classified DGA as a disease, suggesting game industry stakeholders diverted the conversation post-decision. Romashkina and Khuziakmetov [21] highlighted the lack of parental awareness concerning internet addiction risks for children. Several other studies [22], [23], [24] provided varied perspectives ranging from story narration in digital games to the relation between drug addiction and game-based materials. Hu and Zhang [25] narrated their experiences with game addiction through an autoethnographic method, sharing their journey

of overcoming addiction. Lee et al. [26] explored how digital and internet game addictions affected the empathy levels of nursing students in distance education, emphasizing the importance of digital empathy. Cheah et al. [27] conducted a systematic literature review on players' motivations to play digital games, identifying six motivational themes and later proposing theoretical and methodological advancements to study the link between players' motivations and the effects of digital games. Chatterjee et al. [28] proposed a theoretical model based on several theories (social exchange, social network, motivation, and technology acceptance) to investigate the causes and consequences of mobile game addiction, later testing this model with data from mobile game players. Elkhateeb et al. [29] highlighted the detrimental impacts of gaming through various digital platforms on individual and societal health while also discussing the causes and consequences of game addiction. Alan et al. [30] proposed a conceptual model explaining the intent to replay games, factoring in the role of control focus, flow experiences, and outcome expectations, offering various insights and suggestions for the gaming sector and research. Kaimara et al. [31] focused on children's development and the impacts of virtual reality technologies in gaming environments, showcasing both positive and negative effects on physical, cognitive, and psychosocial domains. Genis and Ayaz-Alkaya [32] carried out a cross-sectional study to determine DGA, social anxiety, and parental attitudes among adolescents, using four different scales to collect data from a sample of 1379 students aged 14-17.

Research in the field of software and technology, particularly in social informatics, has been conducted since the 2010s. Alrobai et al. [33] emphasized the need to consider DA in software development. Ali et al. [34] advocated for software options to counter addictive use and noted the industry's lack of awareness about DA's consequences. Alrobai et al. [35] developed a technique to unite people against DGA using online peer support groups, though adapting this for highly addicted individuals proved challenging. Ali [36] discussed the inclusion of the software engineering community in promoting conscious tech use, while Hazar and Halisdemir [37] found a positive relationship between DGA and motivation, observing lower DGA in licensed athletes. Aldhayan et al. [38] examined the factors affecting the acceptance and refusal of online peer support groups for digitally addicted individuals. Ruedas et al. [39] employed discussion groups and in-depth interview techniques to deduce that emerging technology addictions can lead to value loss and bullying. Bhagat et al. [40] explored the effects of online social interaction, online anonymity, and users' need for social interaction on DGA. Wijanarko and Wijanarko Hadisumarto [41] linked the growing revenues of game publishers and rising competition with heightened DGA.

While facial recognition techniques could potentially enhance the detection of digital addiction by analyzing behavioral and emotional cues, their implementation was

excluded from this study due to ethical considerations, privacy concerns, and dataset constraints, as such methods require a specialized dataset and a distinct methodological framework beyond the scope of this research.

In recent years, studies have focused on other complex factors, including mental health and culture. Li et al. [42] employed Mendelian randomization to explore the impact of computer games on mental health using genome-wide association study data. Their findings suggested a potential protective effect of computer games on mental health challenges, highlighting the positive aspects of computer gaming and providing insights into digital intervention strategies for health enhancement. Taylor [43] analyzed the historical and socio-cultural contexts of DAs, viewing them not just as individual problems but as part of a social diagnostic process. Gul and Gul [44] delved into the relationships between attention deficit hyperactivity disorder, slow cognitive tempo, internet addiction, and internet gaming disorder among medical students and resident doctors, underscoring the potential risk slow cognitive tempo poses for internet addiction and gaming disorders. Chen [45] investigated the use of mobile games as a means of stress relief, focusing on mobile games among adolescent students in China. This study sought to determine which types of mobile games are most effective in eliciting healing experiences and what motivates adolescents to turn to these games for stress relief. Heo and Park [46] addressed a critical aspect of video game consumption by exploring how users' play patterns influence their subsequent game purchases, particularly after expressing their opinions through review systems. Keskin and Aral [47] aimed to elucidate both the positive and negative aspects of digital games from the perspective of adolescents. Their study, which employed a survey with open-ended and multiple-choice questions, collected quantitative data from 106 volunteer adolescents aged 12-14. Kazaz et al. [48] observed that digital gaming negatively impacts students' familial, educational, and social lives. Furthermore, they examined the effects of digital media on education, collecting data categorized by parents, teachers, students, and socio-cultural reflections. Bezençon et al. [49] evaluated Escape Addict, a digital escape room intervention, to test its efficacy in conveying addiction prevention messages. Their findings endorsed digital escape rooms as engaging and informative platforms. Other notable works include contributions by Davidson et al. [50] on the mass adoption of digital technologies and their societal effects, Miranda et al. [51] introducing the concept of "Shining Game Design Patterns," and Lindenberg and Basten [52] integrating neurocognitive and clinical psychological perspectives on the effects of digital games. Several more studies provide valuable insights into the digital gaming landscape and its implications. Kaya and Pazarcikci [53] pointed out that DGA is a serious issue that needs to be studied. They examined the factors of DGA by researching on adolescents. A review of the research literature reveals a wide variety of studies on DGA that hold great importance. The diverse body of

literature on DGA highlights its multifaceted nature, encompassing psychological, social, and cultural dimensions while emphasizing the critical role of technology in understanding and addressing this growing issue.

From a methodological perspective, the literature presents numerous studies on detecting DGA focused on identifying DGA through different types of research including qualitative, quantitative, or experimental research. These studies commonly employ psychological tests and surveys [14], [32], [54], [55], [56], [57], interviews [5], [12], [21], [39], [58], observations [55], [56], [59], and experimental methods [8], [12], [14] to gather data. In recent years, machine learning models have been applied by researchers to address the escalating issue of DGA. A methodological literature review was conducted to analyze textual data and identify patterns associated with DGA [3].

The literature includes methodological reviews of studies conducted in the field of addiction. Morgando et al. [60] summarized machine learning methods used in addiction research. Bouhadja and Bouramoul [61] reviewed the latest applications of machine learning in addressing addiction disorders, covering 21 selected works focused on substance and non-substance addictions. Their study divided the research into prognosis, diagnosis, and predicting treatment success, highlighting opportunities for applying new deep learning technologies in addiction research. Wilkinson et al. [62] reviewed the integration of computational methods to enhance the understanding and treatment of substance use disorders, including algorithmic models, dimensionality reduction, and machine learning methods in advancing addiction neuroscience.

In addition to literature research, there are also practical applications using machine learning in the field of addiction. Abada and Bouramoul [63] predicted the risk of substance abuse, specifically drug addiction, using a Naïve Bayes algorithm and achieved an accuracy score of 91.4%. Drug addiction involves excessive use of substances like alcohol, cannabis, cocaine, or heroin, and can manifest as physical or psychological dependence. Diagnosing it is complex, requiring an analysis of the addict's behavior, environment, and family relationships. DA has similar consequences to other types of addiction. Yang et al. [58] evaluated the feasibility of applying supervised machine learning to conduct large-scale content analysis of an online peer-to-peer discussion forum to discover substance addiction, using regression analysis. Sujana et al. [64] analyzed a drug addiction dataset from Reddit, applying models such as Random Forest, Logistic Regression, SVM, KNN, MLP, RNN, and CNN. Their results indicated that the Multi-Layer Perceptron (MLP) model achieved the highest accuracy of 77%, outperforming traditional machine learning models. Kumara et al. [65] focused on predicting drug addiction rates, identifying types of drugs prone to addiction, assessing long-term addiction potential, and predicting relapse after rehabilitation by utilizing Convolutional Neural Networks for image processing, achieving

91.09% accuracy in identifying drug addicts. They also used Random Forest and Decision Tree classifiers for other predictions, showing high accuracy in various tasks, including predicting addiction rates and identifying associated diseases and treatments. This study highlights the potential of machine learning in improving drug addiction prevention, detection, and treatment strategies.

Numerous studies have been conducted on digital addiction, aiming to define its core characteristics and develop effective assessment frameworks. Almourad et al. [2] provided a comprehensive review of digital addiction, identifying key features such as excessive engagement, compulsive behavior, withdrawal symptoms, and significant negative consequences on daily life. Their study emphasized the need for a standardized definition to enhance diagnostic accuracy and intervention strategies. By consolidating various perspectives from the literature, their work contributes to a more structured understanding of digital addiction, supporting the development of automated detection and prevention mechanisms. Studies collectively address various aspects of DGA, providing insights into its causes, impacts, and potential solutions across different demographics and contexts. Singh and Babbar [66] utilized Bayesian networks to detect Internet Addiction Disorder (IAD), demonstrating the effectiveness of probabilistic graphical models in identifying complex behavioral patterns associated with addiction. Their research highlighted the potential of machine learning techniques, such as Bayesian networks, in analyzing user behavior data to predict and diagnose IAD, offering a foundation for developing automated, data-driven intervention strategies. Bayesian Networks were utilized to develop three distinct models for detecting Internet Addiction Disorder (IAD): IAT-BN, CIUS-BN, and PIUQ-BN. The reported accuracy rates for these models were 100%, 95%, and 90%, respectively. Machine learning models have also been used in the field of DGA in recent years. Du [67] analyzed game data to predict user trends for the word game "World." The study used descriptive statistical analysis, ARIMA models, and decision tree models to forecast user numbers and game development prospects. The findings indicated that the game maintains a stable user base, and the decision tree model accurately predicts word difficulty with an accuracy rate of 80%. Ural et al. [3] used the CHAID decision tree algorithm to investigate the relationship between DGA and eating behavior and sleep disorders in children by collecting 252 survey responses. Tusher et al. [68] investigated online gaming addiction using different machine learning classification algorithms with data collected from over 401 data points. The algorithms included Decision Tree, Random Forest, Multinomial Naive Bayes, Extreme Gradient Boosting, SVM, and KNN, with the Multinomial Naive Bayes algorithm demonstrating the highest accuracy. Kornev et al. [69] discussed the development of a machine learning-based platform designed to predict gaming behavior using neuroimaging data. The research focused on understanding the

correlation between gaming activities, decision-making, and brain activation. Several machine learning models, including Multiple Regression, Classification and Regression Trees, Artificial Neural Networks, SVM, and Random Forest, were trained to predict gaming behavior. The results showed that the SVM model with a radial basis function kernel provided the best prediction accuracy, indicating its potential utility in diagnosing and understanding gaming disorders. The study highlighted the effectiveness of combining cognitive tasks with advanced neuroimaging and machine learning techniques to predict behavioral patterns and brain responses related to gaming addiction. Güllü et al. [70] investigated the relationship between obesity, physical activity, and DGA among adolescents, utilizing machine learning to predict DGA. Data from 405 adolescents in Turkey were analyzed, revealing that higher physical activity levels reduce the risk of obesity and DGA. Kuna [71] summarized the application of machine learning techniques to predict the popularity of video games by evaluating Logistic Regression, Decision Tree, and Random Forest, using a dataset from Kaggle. The Random Forest model emerged as the most accurate, with an accuracy of 89.38%. Juwita et al. [72] evaluated the effectiveness of various data mining algorithms in classifying internet gaming disorder, a concept focused on gaming behavior. They compared models such as Random Forest, Decision Tree, Logistic Regression, and SVM. The findings showed that the Random Forest algorithm outperformed other models with the highest accuracy rate of 91.09%.

With the advent of big data, the analysis of unstructured data and the application of NLP in various languages have begun to gain traction in the field. Kumara et al. [65] leveraged machine learning models, particularly focusing on processing unstructured data, to predict individuals' susceptibility to DGA based on short text inputs. Choi and Seo [73] explored patterns and trends in internet gambling by employing text mining techniques, analyzing large volumes of text data from various sources related to internet gambling to identify key themes, sentiments, and patterns. They used NLP to present their findings. Despite growing scholarly attention to DGA, researchers continue to face significant methodological and theoretical challenges. Among the most prominent issues are inconsistent diagnostic criteria, limited dataset diversity, cultural variations in gaming behavior, and the need for automated early detection systems. Furthermore, while numerous machine learning approaches have been proposed, their generalizability and real-world applicability remain limited. To contextualize our study within the current state of research, Table 1 summarizes key studies, highlighting methodologies, datasets, and primary contributions. Among the studies reviewed, a substantial portion directly aligns with the focus of our work, including the application of machine learning and data-driven approaches for detecting and understanding DGA [28] through user behavior and psychological profiling. However, several references (e.g., [1], [7], [9], [23], [24], [26], [27], [37], [42], [49], [51], [54], [55].) were identified as less relevant, as they either emphasize conceptual

discussions, game design ethics, alternative behavioral outcomes, or use non-textual or non-computational methods unrelated to ML-based DGA classification or prediction.

Our study distinguishes itself from the existing literature by focusing on short-text user comments to perform automatic classification for early detection and intervention of digital game addiction. Unlike prior research relying primarily on survey instruments or scale-based methods, our approach leverages natural language data to make real-time predictions. We compared the performance of more than five machine learning models and developed an interface based on the best-performing algorithm.

Various machine learning models have been employed in studies conducted in the field of addiction. In substance addiction research, models such as Naïve Bayes, regression analysis, Random Forest, Logistic Regression, SVM, KNN, Multi-Layer Perceptron (MLP), Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), and Decision Trees have been utilized. These studies aimed to predict addiction risk, forecast treatment success, and analyze behavioral data related to substance addiction. In the field of DGA, researchers used various machine learning models to analyze game data, predict user trends, and understand gaming behaviors. These models include the CHAID decision tree algorithm, Multinomial Naive Bayes, Decision Trees, Random Forest, Multiple Regression, Artificial Neural Networks, SVM, and Logistic Regression. Additionally, text mining and NLP techniques have been employed to predict the popularity of video games and forecast behavioral patterns related to DGA. Specific models include Naïve Bayes for predicting substance abuse risk [63], regression analysis for evaluating supervised machine learning [58], and various models such as Random Forest, Logistic Regression, SVM, KNN, MLP, RNN, and CNN applied to analyze drug addiction datasets [64]. Additionally, Kumara et al. [65] employed CNN for image processing and various classifiers for predicting drug addiction rates and relapse potential. In the realm of DGA, models like the CHAID decision tree [3], Multinomial Naive Bayes [68], and SVM with a radial basis function kernel [69] have been used. Other studies [70], [71] applied machine learning to understand relationships between physical activity, obesity, and DGA, and predict video game popularity. The use of NLP and text mining techniques was explored [73] further expanding the application of machine learning in addiction research.

In the study, NLP and different machine learning algorithms were comparatively applied, and an innovative solution was developed for the early detection and intervention of DGA. Unlike traditional methods, our study utilized machine learning and NLP techniques to analyze short text inputs related to DGA. Traditional methods (psychological tests, surveys, therapies) are often time-consuming, costly, and rely on subjective self-reports, which have certain limitations. Our approach aimed to detect addiction using machine learning algorithms through objective and measurable analyses.

TABLE 1. Summary of recent studies on digital game addiction (2019–2024).

Reference	Year	Methodology	Dataset	Key contributions
[14]	2020	Development and feasibility study	Adolescents and adults with IGD	Proposed a VR-based intervention to encourage alternative leisure activities for individuals with Internet Gaming Disorder
[28]	2022	Empirical analysis	Mobile game users	Investigates behavioral and psychological factors influencing mobile game addiction
[30]	2022	Exploratory study	Turkish online gamers	Investigates psychological factors in online gaming, focusing on flow and outcome expectations moderated by locus of control
[32]	2023	Cross-sectional study	Adolescents	Examines the correlation between digital game addiction, social anxiety, and parental attitudes among adolescents
[40]	2020	Quantitative study	Online game users	Investigates psychological antecedents such as social interaction needs and interpersonal skills in DGA
[44]	2023	Comparative study	Individuals assessed for SCT, ADHD, and gaming disorder	Finds stronger association of SCT symptoms with internet gaming disorder than ADHD, highlighting a distinct risk factor
[47]	2022	Survey study	Adolescents	Explores adolescents' perceptions of both beneficial and harmful effects of digital games
[53]	2023	Structural Equation Modeling (SEM)	Adolescents (web-based)	Analyzes key risk factors contributing to digital game addiction using SEM
[57]	2016	Empirical survey-based study	Gamers	Explores factors that lead to and result from game addiction
[60]	2022	Review and methods paper	Addiction research (general, with ML focus)	Outlines key machine learning techniques applied in addiction research
[61]	2022	Systematic literature review	Addiction disorders	Presents a comprehensive review of recent ML applications targeting addiction diagnosis and intervention
[72]	2023	Comparative analysis of data mining algorithms	Internet Gaming Disorder dataset	Evaluates and compares various data mining algorithms for classifying IGD

Source: Created by the authors.

Traditional methods, including psychological tests, surveys, and therapies, can be time-consuming and costly. Regular attendance at therapy sessions is required, which entails additional costs and time loss. In contrast, our methodology can quickly and cost-effectively through short text analyses. It allows users to self-assess via an online interface, significantly reducing costs and time. Psychological evaluations in traditional methods typically rely on subjective self-reports, which can affect the accuracy and reliability of responses. Machine learning models and NLP techniques used in our study, however, provide data-driven and objective analyses. In traditional methods like surveys and interviews, data collection depends on participants' subjective

perceptions. Since real-time data is not collected through methods like surveys, responses are perceptual-based on the scale of questions asked. This study worked with unstructured text data. Moreover, psychological counseling and therapies can be limited by physical location and accessibility. Some individuals may have difficulty accessing these services or may avoid them due to social stigma. Our developed online user interface provides a platform that users can access from anywhere and at any time, increasing the potential for wider reach and early intervention.

The accuracy and performance of the machine learning models used in the study were evaluated using various metrics (accuracy, precision, recall, F1 score), and the model with the

best performance was selected. Accordingly, the model with the highest accuracy in predicting addiction tendencies was chosen through comparative analysis. Model performances were measured using various metrics (accuracy, precision, recall, F1 score) to obtain objective and measurable results. In this context, this study aims to provide a new perspective by utilizing artificial intelligence and machine learning techniques in the exploration and understanding of DGA. Our research is based on data obtained from online game feedback, seeking to explore DGA. This study aims to contribute to our understanding and management of DGA.

III. METHOD

The development and increasing prevalence of DGA necessitate sophisticated methods of identification and intervention. To this end, we adopted a novel approach using machine learning to analyze short text inputs to predict individuals' propensity for DGA. Such an approach not only leverages advancements in data analytics but also underscores the significance of unstructured data in generating actionable insights. By harnessing the potential of textual data mining, this study offers an innovative methodology in addiction studies, specifically for DGA. This section details the process of data collection and preprocessing used in this research. Python programming language is used in the study.

To enhance methodological transparency and reproducibility, a structured algorithmic framework is presented. The workflow comprises the following sequential stages:

- **Data Collection:** Acquisition of textual inputs related to gaming behavior.
- **NLP:** Basic linguistic processing, including tokenization and lemmatization.
- **Data Preprocessing:** Noise removal, stopword elimination, and vectorization.
- **Data Augmentation:** Enrichment of data through paraphrasing and semantic variation.
- **Modeling:** Implementation and evaluation of machine learning classifiers for addiction detection.
- **User Feedback:** Integration of user feedback to support iterative model refinement.

This pipeline offers a modular and adaptable foundation for the early detection of digital game addiction through natural language processing and machine learning techniques.

This study integrates standard machine learning components into a unified pipeline specifically tailored for DGA detection. The modular design, combining semantic augmentation and a user feedback loop, represents a novel implementation for early intervention systems and enhances practical applicability.

A. DATA COLLECTION

DGA has become an increasingly serious problem in recent times. The consequences of game addiction can devastate users and their surrounding community. In line with the research objective, the dataset comprises comments from digital game addicts and those who want to highlight the

dangers of this type of addiction, including individuals who have suffered from and recovered from this addiction in the past. The data were collected from various websites between May 1, 2023, and June 1, 2023 [74]:

- *Game Addiction Quotes*¹
- *Addiction Slogans 2023*²
- *Best Video Game Quotes*³
- *Inspirational Quotes About Gaming*⁴

The comments on these websites consist of those made by digital game addicts and users who want to draw attention to DGA. These websites invite users to comment with statements such as, "Playing video games is a normal hobby, and many people, both young and old, are completely engrossed in this hobby. However, some become so absorbed in their gaming obsession that it takes over their entire lives." The data were collected using the web scraping method. To avoid overloading the servers of the websites and violating data extraction rules, the guidelines specified in the websites' robots.txt files were followed. For websites without this file, the standard procedure was applied. The Python programming language was used, and the BeautifulSoup, Selenium, and Regex libraries were utilized. Since the data were spread across many pages, a loop was used to traverse all pages, and the data were extracted according to the classes in the HTML structure. The collected data were saved in CSV format.

DGA has increasingly become a serious issue in recent years, with significant consequences affecting individuals and their surrounding communities. In line with the research objective, the dataset comprises comments from digital game addicts and individuals who aim to highlight the dangers of this type of addiction, including those who have suffered from and recovered from it in the past. The data were collected from various websites between May 1, 2023, and June 1, 2023 [74], including sources such as Game Addiction Quotes, Addiction Slogans 2023, Best Video Game Quotes, and Inspirational Quotes About Gaming. These websites feature user-generated content, including comments written by digital game addicts and individuals who seek to raise awareness about gaming addiction. Users were encouraged to share their perspectives through statements such as: "Playing video games is a normal hobby, and many people, both young and old, are completely engrossed in this hobby. However, some become so absorbed in their gaming obsession that it takes over their entire lives."

The data were collected using web scraping techniques, ensuring adherence to ethical guidelines. To prevent server overload and comply with data extraction policies, the robots.txt files of the websites were consulted. If no such file was available, a standard data extraction procedure was

¹Game Addiction Quotes: <https://sweetlovesmessages.com/game-addiction-quotes/>

²Addiction Slogans 2023: <https://www.bestslogans.com/list-ideas-taglines/gaming-addiction-slogans/>

³Best Video Game Quotes: <https://sweetlovesmessages.com/best-video-game-quotes/>

⁴Inspirational Quotes About Gaming: <https://graciousquotes.com/gaming/>

followed. The Python programming language was utilized for data collection, employing the BeautifulSoup, Selenium, and Regex libraries. Given that the data were spread across multiple pages, an automated loop was implemented to systematically traverse and extract comments based on the HTML structure. The collected data were then saved in CSV format.

The comments were collected in their original form and manually annotated into two categories: 1 (addiction-related comments), written by digital game addicts, and 0 (non-addiction-related comments), written by individuals who aim to highlight the risks of gaming addiction or those who have overcome it. The original dataset contained 660 textual comments, with 330 comments from digital game addicts and 330 comments from individuals emphasizing the potential dangers of addiction. This balanced dataset ensured representation from both perspectives, forming the foundation for further processing and analysis.

To improve the quality and consistency of the dataset, an expert review process was conducted. Initially, all sentences in the Data (1).csv dataset were labeled as containing gaming addiction-related expressions (VALUE = 1). To balance the dataset, non-addiction-related (VALUE = 0) sentences were generated by modifying the original sentences using negations while maintaining their semantic integrity. During the expert review process, addiction-related sentences (1) were verified, and non-addiction-related sentences (0) were validated. Additionally, sentences that were grammatically incorrect or semantically ambiguous were marked as (−1) and removed. As a result, 245 irrelevant or incoherent sentences were eliminated, increasing the dataset's reliability.

To assess the agreement between the initial annotation and expert evaluation, Cohen's Kappa metric was computed, yielding a score of 0.86, which indicates a high level of consistency between the original labeling and expert validation. Table 2 presents sample sentences from the dataset, demonstrating both addiction-related (1) and non-addiction-related (0) examples. The initial dataset consists of 415 instances, serving as the foundation for subsequent preprocessing and processing steps.

B. DATA AUGMENTATION TECHNIQUES

To augment the dataset, methods such as data augmentation with BERT, synonym replacement, and back translation were employed. The Cosine similarity metric was used for performance evaluation by comparing the source file with the output file. After applying each algorithm, the differentiation of the dataset was manually examined, and comparative data were analyzed using an algorithm with the random library to evaluate changes in the sentences within the new datasets. No corrections were applied manually. The methods used are as follows:

- Data Augmentation with BERT: A total of 830 dataset were generated using the BERT model with data augmentation. The dataset consists of 1,245 records. The Transformers library was utilized. The similarity

between the original and newly generated data were measured using Cosine Similarity. The average similarity score between the original and new comments was found to be 0.9713, indicating that the newly created comments differ significantly from the original ones, likely containing various word and structure changes. A low similarity score suggests that the model generated diverse and innovative comments, demonstrating its creativity and capacity to diversify the dataset.

- Synonym Replacement: To preserve the original comments in the dataset, another data augmentation method, synonym replacement, was applied in sequence. Using the NLTK's WordNet lexical database, the dataset was expanded by replacing some words in each sentence with their synonyms. To ensure meaningful substitutions, SpaCy's NLP pipeline was used to tokenize the text and extract part-of-speech (POS) tags, specifically focusing on nouns, verbs, and adjectives while excluding stopwords from the NLTK stopword corpus. The Cosine similarity performance result was 0.0648. After synonym replacement, the dataset size increased to a total new dataset size of 3,320. The low similarity score between the original and augmented datasets indicates significant differences due to various word and structure changes, suggesting that the model created diverse and innovative comments, enhancing the dataset's variety for model training.
- Back Translation: The method utilizes the deep-translator library to translate English text into Turkish and then back into English, introducing lexical and syntactic variations while preserving the original meaning. The dataset is processed using pandas, ensuring proper handling of missing values to maintain data integrity. Additionally, robust error-handling mechanisms are implemented to mitigate potential API failures, incorporating automatic retries and controlled delays (*time.sleep()*) to prevent exceeding API rate limits. The dataset consists of 3,320 original records, which, following back translation, results in a total of 6,640 records, effectively expanding the dataset while maintaining semantic consistency. This diversity is beneficial for training machine learning models, as it improves their generalization capability. The low similarity score demonstrates the effectiveness of back translation in creating a varied dataset, enhancing the model's robustness to different expressions and language constructs. This process introduced considerable variation in word choice and sentence structure, resulting in a diverse and innovative augmented dataset.

The final dataset contains 6,640 rows.

C. DATA PREPROCESSING

Data preprocessing is undeniably a pivotal step in data analysis and machine learning. Regardless of the performance prowess of the applied algorithms, without a meticulously executed preprocessing phase, the results tend to

TABLE 2. Sample of the dataset.

Comments	Value
This game is the only thing that makes me cry and laugh	1
This game is more than an addiction. It's a necessity.	1
This game is my dream. It's my fantasy and my imagination comes to life.	1
Playing digital games can be enjoyable and beneficial in moderation.	0
Digital addiction is not a static condition but a dynamic and progressive problem.	0
Digital addiction is not your friend. It is your enemy.	0

underperform. This phase aims to transform raw data into a format that is both meaningful and efficient for subsequent analysis [75].

Data preprocessing ensured that the dataset was prepared for further modeling steps, with improved text quality, reduced noise, and enhanced contextual understanding. By rigorously these preprocessing steps and documenting the process, we ensure that the data is in an optimal state for subsequent machine learning analysis. Kahya Ozyirmidokuz and Celik [76] presented algorithms for the data preprocessing steps. The libraries used are as follows: Pandas, String, Matplotlib, NLTK, SpaCy, NegspaCy, and Scikit-learn. The distribution of word counts before and after cleaning is shown in Figure. In the data preprocessing phase, several key steps were taken to clean and prepare the dataset for analysis:

- The data preprocessing process was carried out using the Pandas library to ensure data integrity and reliability. Initially, the dataset was examined for missing values in the “EXPERT_VALUE” column, and no missing values were detected, eliminating the need for imputation or removal of records. Subsequently, duplicate records were identified and removed, resulting in the elimination of 2,988 duplicate rows. This process ensures that the dataset contains only unique and complete entries, minimizing redundancy and improving data quality for subsequent analysis. The total dataset consists of 3,652 records.
- Data Cleaning: One primary aspect of our preprocessing was data cleaning. This entailed eliminating or correcting missing, corrupted, or inconsistent data. Often, missing data can be imputed using statistical methods such as mean, median, or mode. In our analysis, a notable component of this process was the removal of punctuation and converting all characters to lowercase. This was achieved using Python’s string manipulation capabilities, ensuring that the algorithms could learn effectively without unnecessary distractions. The ‘COMMENT’ column, which contained textual data, underwent several cleaning steps:
- Case Normalization: All text was converted to lowercase using Python’s string methods.
- Punctuation Removal: Punctuation was removed from the text using the string library.

- Stopwords Removal: The next significant step was the removal of stopwords from the dataset. Even though there exist standard lists of stopwords in the literature, our study enhanced and used an advanced version to suit the dataset’s nuances. This step aligns with the general preprocessing guideline of data transformation, which helps in making the data ready for analysis. We utilized the Natural Language Toolkit (NLTK) library in Python to accomplish this task.
- Negation Handling: Recognizing the complexities of handling negation within textual data, we integrated an advanced negation detection and transformation mechanism to enhance contextual understanding. This process was implemented using the spaCy library and the Negex pipeline, which allows for rule-based negation detection within the dataset. The approach involved identifying negated words in the text and prefixing them with “NEG_”, ensuring that their contextual meaning remains distinguishable during further analysis. A total of 9,520 comments have been processed, where negated terms were appropriately transformed. The negation handling framework included terms commonly associated with negation, i.e., [“no”, “not”, “n’t”, “never”, “none”]. These words were dynamically detected, and the corresponding tokens were annotated with a “NEG_” prefix to explicitly mark their negated status. This modification helps in preserving the semantic integrity of negated expressions, improving the accuracy of downstream text processing tasks.
- N-grams Creation: To better capture context, we incorporated n-grams, particularly bigrams (2-grams) and trigrams (3-grams). An n-gram is a contiguous sequence of n items from a given sample of text or speech. Using n-grams helps in understanding the context of words within the text more accurately. The Scikit-learn library in Python facilitated this process, enabling the creation of n-grams from the text data. While the original text data were analyzed on a word-by-word basis, n-grams allowed for the creation of more complex and meaningful word groups. Using n-grams, we obtained 5000 features that indicate the frequency of specific word groups. Consequently, the new dataset contains more meaningful and contextually relevant word groups compared to the original dataset.

The application of n-gram processing has significantly influenced the structure of the text data, as demonstrated in Figure 2, which illustrates the word count distribution before and after processing. The histogram, generated using Matplotlib and Seaborn, visually represents the impact of data preprocessing on the distribution of word counts. Notably, the word count has generally increased for many comments, as indicated by the rightward shift in distribution. While the most frequent word count range remains between 5 and 15 words, there is a discernible rise in the number of longer comments post-processing, reflecting the formation of extended token sequences due to n-gram extraction.

The final dataset now consisted of 2,594 rows, incorporating 20,905 additional attributes generated through n-gram feature extraction, alongside the COMMENT and VALUE attributes. This refined dataset was now structured for advanced NLP techniques, facilitating further computational analysis and meaningful insights.

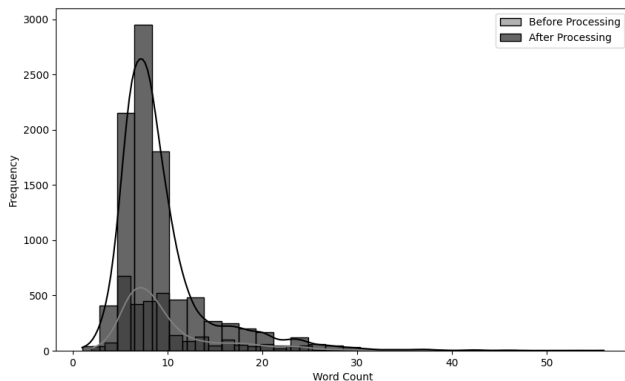


FIGURE 2. Word count distribution before and after processing.

D. CLASSIFICATION MODEL APPLICATIONS

Following the data preprocessing steps, the dataset was prepared for direct utilization in various classification algorithms. To extract meaningful numerical representations of the text data, TF-IDF (Term Frequency-Inverse Document Frequency) vectorization from the Sklearn.feature_extraction.text module was employed, with optimized feature selection to prevent overfitting. Feature reduction techniques were applied, limiting the vocabulary size (`max_features=1000`) and removing rare words (`min_df=2`) to enhance model generalization. The classification models were trained using balanced class weighting, depth constraints, and hyperparameter tuning to ensure optimal performance. The details of the models are as follows:

- **Logistic Regression:** Logistic Regression is a statistical classification method that models the probability of a given class label based on input features. It was implemented using the `LogisticRegressionCV` module from Scikit-learn, incorporating TF-IDF vectorization to convert text data into numerical features. To mitigate

overfitting, stronger L2 regularization ($C_s=[0.001, 0.01, 0.1, 1]$) and balanced class weighting were applied. The model was trained using 10-fold cross-validation, ensuring that it generalized well to unseen data.

- **K-Nearest Neighbors (KNN):** KNN is a non-parametric algorithm that classifies instances based on the most frequent label among their closest neighbors. The Scikit-learn implementation was used, and TF-IDF transformation was performed on the textual data. To reduce overfitting, feature selection constraints (`max_features=1000`) and distance-based weighting (`weights="distance"`) were applied. `GridSearchCV` was utilized to find the optimal number of neighbors (`k`), distance metrics, and weighting functions. Stratified cross-validation was used to ensure robust evaluation.
- **Support Vector Machines (SVM):** The SVM model, implemented using `SVC` from Scikit-learn, was optimized for robust generalization by controlling the complexity of the decision boundary. A linear kernel was chosen to maintain model interpretability, while hyperparameter tuning was performed to adjust the regularization parameter ($C=0.1$) and ensure balanced training. This approach helped prevent overfitting while preserving classification accuracy.
- **Decision Trees:** Decision Trees split datasets into subsets based on information gain, making them interpretable yet prone to overfitting. The Scikit-learn implementation of `DecisionTreeClassifier` was used with maximum depth constraints (`max_depth=10`), minimum samples per split (`min_samples_split=5`), and balanced class weighting. Hyperparameter tuning via `GridSearchCV` with 10-fold cross-validation was applied to optimize tree complexity while ensuring generalization.
- **Random Forests:** As an ensemble learning method, Random Forests construct multiple decision trees to improve accuracy and reduce overfitting. The Scikit-learn implementation of `RandomForestClassifier` was employed, with `n_estimators` limited to 200, maximum tree depth set to 15, and minimum samples per split set to 5. This setup ensured that trees were not excessively deep, preventing memorization of training samples. Class weighting and cross-validation techniques further improved model robustness.
- **XGBoost:** XGBoost is an optimized gradient boosting algorithm that sequentially trains weak learners to minimize errors. The model was implemented using `XGBClassifier`, incorporating stronger regularization (`reg_lambda=1`, `reg_alpha=0.5`), subsampling techniques (`subsample=0.8`, `colsample_bytree=0.8`), and early stopping to prevent overfitting. The number of boosting rounds was limited (`n_estimators=200`), and learning rate tuning (`learning_rate=0.1`) was applied to ensure gradual convergence without excessive complexity.

In our research, we observed that the model's performance metrics were exceptionally high, indicating a high likelihood of overfitting. Consequently, we employed the early stopping technique to halt the training once the model reached a certain performance threshold. This adjustment aimed to prevent the model from overfitting and to ensure more generalized learning. Since KNN does not support early stopping natively, we relied on cross-validation and careful hyperparameter tuning to control overfitting. To address overfitting in the Decision Tree model, techniques such as setting constraints on tree depth, minimum samples required for splits and leaves, and performing robust cross-validation were employed. For the Random Forest model, constraints on hyperparameters and robust cross-validation were used to prevent overfitting. Although Random Forests are generally less prone to overfitting compared to single decision trees, careful tuning of parameters such as the number of estimators, maximum depth, minimum samples required to split an internal node, and minimum samples required at a leaf node is crucial. Table 3 summarizes the models, the libraries used, and the hyperparameters that were tuned for each model. To mitigate overfitting and enhance the generalization of the models, several modifications were applied across all classification approaches. Feature reduction in TF-IDF vectorization was implemented by limiting the vocabulary size to a maximum of 1,000 features (`max_features=1000`) and removing infrequent words using a minimum document frequency threshold (`min_df=2`). Additionally, bigram features (`ngram_range=(1,2)`) were selectively incorporated to retain meaningful word associations while preventing excessive feature expansion. To address class imbalance, `class_weight="balanced"` was incorporated into models that support it, such as Logistic Regression, Decision Trees, Random Forests, and SVM. In cases where class imbalance was more pronounced, Synthetic Minority Over-sampling Technique (SMOTE) was considered to ensure more balanced training data. Regularization techniques were applied across models to further reduce overfitting, including stronger L2 regularization ($Cs = [0.001, 0.01, 0.1, 1]$) in Logistic Regression, L1 & L2 penalties (`reg_lambda=1`, `reg_alpha=0.5`) and early stopping in XGBoost, and constraints on tree depth (`max_depth=10-15`) and minimum split size (`min_samples_split=5`) for Decision Trees and Random Forests. Hyperparameter tuning was conducted using GridSearchCV with 10-fold cross-validation to optimize model parameters, ensuring an appropriate trade-off between model complexity and performance. Regularization tuning (C values) was specifically performed for Logistic Regression and SVM, while Decision Trees and Random Forests were optimized through tree depth control. Finally, robust cross-validation techniques were applied, with 10-fold cross-validation conducted across all models, and performance was systematically evaluated on unseen test data to ensure strong generalization and reduced overfitting.

Each model's performance was subsequently evaluated based on various metrics such as accuracy, precision, recall,

and F1-score. Detailed performance metrics and their comparisons, in line with the research objectives, will be covered in the subsequent sections. After selecting the best model, it is necessary to export the model results for further integration.

Using the pickle library, we have ensured that the model and vectorizer are saved and can be easily loaded for prediction purposes on the web application. This enables the web interface to provide results based on the trained model outputs. The exported models were integrated into a minimally designed web page. This interface comprises a text box for user input, a button to initiate the prediction, and a label where the results are displayed.

The system works as follows. Individuals suspecting DGA can write a few sentences about their digital gaming habits, and the web page predicts their tendency towards this addiction. The website will then analyze the input and provide feedback on whether the user shows signs of DGA. Furthermore, the website will provide sentiment cluster results to the users. We also collect the subjective opinions of the users on whether they consider themselves digitally addicted, the extent of their addiction, and their peers' perceptions of their addiction. Additionally, demographic data such as age, gender, and city of residence are collected to provide preliminary diagnostic information. A database was created in MySQL.

E. SENTIMENT ANALYSIS

Sentiment analysis was applied to the 'value' attribute to rate DA. The value attribute in the original dataset holds binary values, where '1' indicates DA and '0' represents awareness of or a warning against DA. These binary values have been diversified into a 0-4 scale to provide more granularity. Each comment's sentiment was analyzed and classified into values ranging from 0 to 4. A value of 0 represents a "Very Negative" sentiment, 1 represents a "Negative" sentiment, 2 represents a "Neutral" sentiment, 3 represents a "Positive" sentiment, and 4 represents a "Very Positive" sentiment. This clustering was accomplished by analyzing each sentence and categorizing sentiment scores according to specific threshold values. The TextBlob library was utilized for this process. This method allows for a more detailed and diversified representation of sentiment within the dataset, enhancing the understanding of the varying degrees of sentiment in the comments, which was not possible with the original binary values. Figure 3 presents the distribution of sentiment scores, showing that the sentiment scores are predominantly centered around 2, indicating most comments have a neutral sentiment. There is a significant peak at 2. Scores of 1 and 3 are moderately frequent, suggesting some slightly negative and positive comments. Scores of 0 and 4 are rare, indicating very few comments are extremely negative or positive. While a Silhouette Score of 1.0 indicates perfect clustering, it is essential to validate the model with additional metrics, cross-validation, and real-world testing. This ensures that the high score does not come at the expense of generalizability, diversity, and practical applicability. Therefore, hyperparameter

TABLE 3. Summary of models, libraries, and hyperparameters.

Model	Library/Implementation	Hyperparameters
Logistic Regression	LogisticRegressionCV from Scikit-learn	Cs=[0.001, 0.01, 0.1, 1], solver='saga', class_weight='balanced'
KNN	KNeighborsClassifier from Scikit-learn	n_neighbors=[3,5,7,9], weights='distance', metric=['euclidean', 'manhattan']
Decision Tree	DecisionTreeClassifier from Sklearn.tree	max_depth=10, min_samples_split=5, class_weight='balanced'
Random Forest	RandomForestClassifier from Scikit-learn	n_estimators=200, max_depth=15, min_samples_split=5, class_weight='balanced'
XGBoost	XGBClassifier from XGBoost	n_estimators=200, max_depth=6, learning_rate=0.1, reg_lambda=1, reg_alpha=0.5, subsample=0.8
SVM	SVC from Scikit-learn	C=0.1, kernel='linear', class_weight='balanced'

Source: Created by the authors.

tuning was performed after sentiment analysis. This approach is common, especially when data needs to be optimized for clustering or classification. Sentiment analysis captures the emotional nuances within the data, making it more meaningful. Consequently, following this step with hyperparameter optimization is logical as it enhances the understanding of the data’s features and improves model performance. The Silhouette Score was determined to be 1.0, indicating highly effective clustering. Hyperparameter tuning was conducted during the modeling phase to ensure optimal performance and accuracy of the sentiment analysis and clustering.

In conclusion, our dataset comprised 2348 comments with five sentiment clusters.

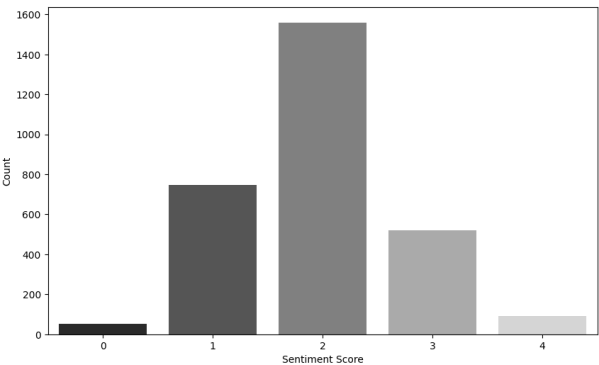


FIGURE 3. Distribution of sentiment clusters.

IV. MODELING RESULTS

A. BINARY LABEL MODEL APPLICATIONS

Table 4 presents the performance scores of the classification models after hyperparameter tuning. These results provide a comprehensive evaluation of each model in terms of accuracy, precision, recall, and F1-score, highlighting their effectiveness in classification tasks. The XGBoost model achieved

the highest performance, with an accuracy of 93.06%, precision, recall, and F1-score all at 0.93. The confusion matrix indicates only 21 false negatives and 10 false positives, demonstrating the model’s superior ability to balance precision and recall while maintaining strong predictive power. The Random Forest model followed closely, achieving an accuracy of 89.21%, with precision, recall, and F1-score all at 0.89. The confusion matrix shows 29 false negatives and 24 false positives, suggesting that while the model effectively distinguishes between classes, it has a slightly higher misclassification rate compared to XGBoost. The Logistic Regression model performed well, with an accuracy of 85.74%, precision of 0.90, recall of 0.86, and an F1-score of 0.87. The confusion matrix reveals 65 false negatives and 9 false positives, indicating that the model is slightly biased toward predicting the positive class but maintains a strong balance overall. The KNN model achieved an accuracy of 80.92%, with precision of 0.79, recall of 0.81, and an F1-score of 0.74. The confusion matrix shows 79 false negatives and 30 false positives, highlighting that while KNN effectively classifies most instances, it struggles with some positive cases. The Decision Tree model achieved an accuracy of 79.58%, with precision, recall, and F1-score at 0.86, 0.80, and 0.81, respectively. The confusion matrix indicates 92 false negatives and 52 false positives, suggesting that while decision trees provide interpretability, they are more prone to overfitting compared to ensemble models. The SVM model demonstrated the lowest performance, with an accuracy of 74.57%, precision of 0.86, recall of 0.75, and an F1-score of 0.77. The confusion matrix shows 120 false negatives and 58 false positives, indicating that the model struggles with recall, potentially leading to a higher rate of misclassified positive instances. 10-fold cross-validation results confirm the stability of the models, with XGBoost and Random Forest exhibiting superior generalization capabilities. Among all models, XGBoost outperformed the others,

TABLE 4. Performance scores of the algorithms after hyperparameter tuning.

Algorithm Type	Accuracy	Precision	Recall	F1-score	Confusion Matrix
Logistic Regression	0,8574	0,9	0,86	0,87	[[93, 9], [65, 352]]
KNN	0,8092	0,79	0,81	0,74	[[72, 30], [79, 338]]
SVM	0,7457	0,86	0,75	0,77	[[44, 58], [120, 297]]
Decision Tree	0,7958	0,86	0,8	0,81	[[50, 52], [92, 325]]
Random Forest	0,8921	0,89	0,89	0,89	[[78, 24], [29, 388]]
XGBoost	0,9306	0,93	0,93	0,93	[[92, 10], [21, 396]]

Source: Created by the authors.

making it the most suitable choice for this classification task. Logistic Regression and Random Forest also delivered competitive performance, making them viable alternatives depending on the application requirements. KNN, Decision Trees, and SVM models exhibited lower predictive performance, suggesting that further optimization or alternative modeling approaches may be needed.

B. EXPANDED DATASET MODEL APPLICATIONS

Table 5 presents the performance scores of the classification models after hyperparameter tuning for the dataset. These findings provide insight into the efficacy of each model in terms of accuracy, precision, recall, and F1-score. The XGBoost model emerged as the best-performing algorithm, achieving an accuracy of 92.49%. The precision, recall, and F1-score are all approximately 0.92, indicating a well-balanced performance. The confusion matrix shows only 7 false negatives and 32 false positives, demonstrating the model's capability to generalize effectively with minimal misclassifications. The Random Forest model followed closely, achieving an accuracy of 91.14%. The precision, recall, and F1-score are all around 0.91, confirming its strong predictive power. The confusion matrix indicates 25 false negatives and 21 false positives, suggesting that while Random Forest provides robust classification performance, it is slightly more prone to false negatives than XGBoost.

The Logistic Regression model achieved an accuracy of 88.44%, with precision of 0.91 and recall of 0.88. The F1-score of 0.89 indicates a strong balance between precision and recall. The confusion matrix reveals 51 false negatives and 9 false positives, suggesting a slight bias toward predicting the negative class. The KNN model achieved an accuracy of 88.63%, with a precision of 0.88 and recall of 0.89. The confusion matrix shows 15 false negatives and 44 false positives, highlighting its tendency to misclassify some positive instances. The Decision Tree model obtained an accuracy of 83.04%, with precision, recall, and F1-score around 0.83–0.84. The confusion matrix shows 79 false negatives and 9 false positives, indicating that the model is prone to overfitting and may require further pruning or regularization. The SVM model had the lowest performance, with an accuracy of 76.69%. The precision, recall, and

F1-score are all around 0.77–0.79. The confusion matrix shows 106 false negatives and 15 false positives, indicating a significant challenge in correctly identifying certain class labels, particularly for positive instances. The 10-fold cross-validation confirmed the stability of the models, with XGBoost and Random Forest demonstrating superior generalization capabilities. Among all models, XGBoost outperformed the others, making it the most suitable choice for this classification task. Logistic Regression, Random Forests, and KNN also provided competitive performance, making them viable alternatives depending on specific application requirements. The Decision Tree and SVM models exhibited lower predictive performance, suggesting the need for further optimization or alternative modeling approaches.

Comparing the results between Table 4 (Binary Labels) and Table 5 (Expanded Dataset), a notable improvement is observed across all models in precision, recall, and F1-score. This suggests that expanding the dataset and introducing additional feature representations improved the model's ability to distinguish between classes, particularly reducing misclassifications. XGBoost and Random Forest saw minor accuracy improvements, but their generalization ability was further reinforced, as indicated by stable precision-recall balances. Logistic Regression and KNN showed significant gains in recall, highlighting improved recognition of class instances that were previously harder to classify. Decision Trees and SVM, while still underperforming relative to other models, benefited from feature expansion, indicating that richer textual representations mitigated some of their previous limitations. Examining the confusion matrices, XGBoost exhibited the lowest number of false negatives across both datasets, reinforcing its strength in correctly classifying instances. Random Forest followed closely, with slightly higher misclassification rates, particularly false positives, suggesting that while its ensemble learning reduces variance, it still faces minor classification boundary challenges. SVM and KNN struggled in correctly classifying negative instances, as evidenced by their high false negative counts. Decision Trees, despite improvements, remained vulnerable to overfitting-induced misclassification errors, particularly in cases where tree depth was not optimally constrained. These findings confirm that

TABLE 5. Performance scores of the algorithms after hyperparameter tuning.

Algorithm Type	Accuracy	Precision	Recall	F1-score	Confusion Matrix
Logistic Regression	0,8844	0,91	0,88	0,89	[[97, 9], [51, 362]]
KNN	0,8863	0,88	0,89	0,88	[[62, 44], [15, 398]]
SVM	0,7669	0,85	0,77	0,79	[[91, 15], [106, 307]]
Decision Tree	0,8304	0,89	0,83	0,84	[[97, 9], [79, 334]]
Random Forest	0,9114	0,91	0,91	0,91	[[85, 21], [25, 388]]
XGBoost	0,9249	0,92	0,92	0,92	[[74, 32], [7, 406]]

Source: Created by the authors.

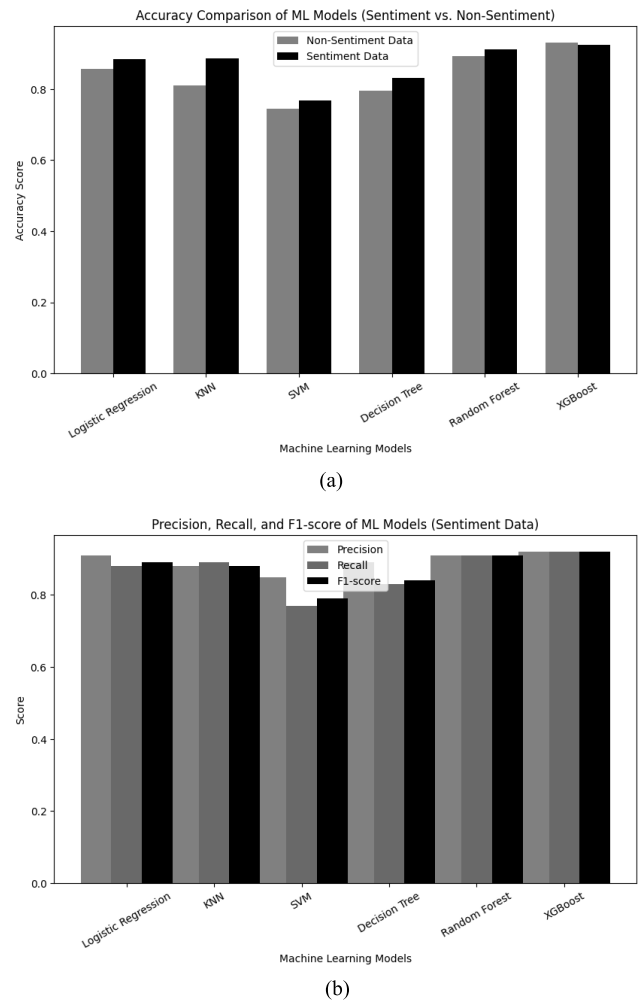


FIGURE 4. Accuracy comparison of machine learning models.

XGBoost is the most suitable model for text classification in this study, given its consistent high performance and minimal misclassification rates. Figure 4 illustrate the comparative performance of various machine learning models on sentiment-augmented and non-sentiment datasets. The results

demonstrate that ensemble learning techniques, particularly XGBoost and Random Forest, consistently achieve higher accuracy and balanced performance across evaluation metrics. Notably, models trained on sentiment-augmented data generally exhibit improved classification performance, suggesting that incorporating sentiment features contributes to better predictive capabilities in digital addiction classification tasks.

The findings of this study confirm that XGBoost is the most suitable model for text classification in digital addiction detection, achieving 92.49% accuracy, 92% precision, and 92% recall, outperforming previous studies and alternative machine learning models. Compared to earlier studies employing traditional machine learning algorithms, our approach achieved up to 93% accuracy with XGBoost, outperforming models such as Random Forest [65] and SVM [49], [50]. Additionally, Random Forest and Logistic Regression emerge as strong alternatives, particularly in balancing interpretability and accuracy, with 91.14% and 88.44% accuracy scores, respectively. However, KNN (88.63%), Decision Trees (83.04%), and SVM (76.69%) exhibit lower predictive reliability, suggesting challenges in handling class imbalances and complex textual patterns. These results indicate that further optimization or alternative modeling approaches, such as deep learning-based architectures, could be explored to enhance classification performance. The integration of advanced neural network models or hybrid ensemble strategies may further refine digital addiction detection and contribute to more effective AI-driven intervention frameworks.

V. CONCLUSION

In today’s rapidly evolving digital landscape, DGA poses a significant challenge at both individual and societal levels. Machine learning presents an innovative pathway for addressing this issue, offering data-driven insights into addiction behaviors. This study aimed to develop a machine learning-based framework to detect and assess DGA by analyzing textual data from digital game addicts and statements

emphasizing the risks of excessive gaming. Multiple classification algorithms were tested, with SVM demonstrating superior validation metrics and outperforming alternative models. The outstanding cross-validation scores following hyperparameter tuning further underscore the model's reliability and robustness.

The findings of this study pave the way for AI-driven digital addiction assessment and intervention systems. Enhanced versions of such models could be integrated into gaming platforms, mental health applications, and educational systems, enabling early detection of addiction patterns, risk assessments, and personalized intervention strategies. By leveraging machine learning-driven sentiment and behavioral analysis, these systems could foster healthier gaming habits and mitigate the negative consequences of excessive gaming.

To effectively address digital addiction, AI-driven interventions such as digital detox applications, parental control systems, and habit-modifying AI tools have gained prominence. These technologies leverage machine learning for personalized monitoring, real-time content filtering, and adaptive engagement strategies to promote healthier digital habits. Future research should focus on integrating psychological counseling modules, peer-support referrals, and adaptive recommendation engines that dynamically adjust intervention strategies based on user behavior. While these approaches present promising solutions, further empirical validation is required to assess their long-term effectiveness and ethical implications. As this study primarily focuses on addiction classification, developing targeted intervention strategies remains a critical area for future interdisciplinary research.

However, this study has certain limitations. The initial dataset of 415 entries presents a constraint, necessitating further expansion to improve generalizability. Additionally, while model predictions provide valuable insights, they should not be solely relied upon for diagnosing addiction. Incorporating complementary diagnostic tools, such as self-reported surveys, behavioral tracking, and expert evaluations, could provide a holistic assessment of addiction severity. Further research should also investigate psycho-social variables influencing addiction, such as gaming motivations, personality traits, and social engagement patterns, to refine the classification process. Misclassifications could lead to either undue concern or a false sense of security. To mitigate this, the system includes clear disclaimers stating it is not a diagnostic tool. Ethical principles including informed consent, transparency, and data privacy have been considered throughout system development.

While a formal error analysis was not conducted in this study, we acknowledge its potential to uncover limitations in detecting nuanced expressions of digital game addiction. Future iterations of the model will include analysis of misclassified cases (e.g., metaphorical or emotionally neutral comments) to refine prediction accuracy and reduce false detections.

While the initial dataset comprised only 415 samples, we employed data augmentation techniques including BERT-based generation, synonym replacement, and back translation, expanding the dataset to 6,640 comments. Nonetheless, future work will focus on collecting larger, more diverse real-world data to enhance generalizability.

Aligned with the ethos of leading research in digital well-being, this study highlights the interdependence between technology, human behavior, and data-driven solutions. It underscores the importance of leveraging artificial intelligence and NLP to address real-world challenges beyond gaming addiction, extending to broader digital behavior analysis and intervention models.

The findings indicate that machine learning-based DGA detection models can be effectively deployed in real-world applications. Integrating the best-performing models into gaming ecosystems and mental health support tools could enable proactive interventions to mitigate addiction risks. Future implementations should focus on real-time detection systems, where AI models dynamically analyze user interactions and behavioral patterns to provide personalized alerts and adaptive interventions. Additionally, collaborations with digital well-being initiatives, policymakers, and mental health professionals could enhance the societal impact of these models. Although a formal error analysis was not conducted in the current study, we acknowledge its importance in identifying model limitations and understanding misclassification patterns. In future work, we plan to include a detailed error analysis—examining examples of false positives and false negatives—to reveal potential challenges such as the detection of metaphorical expressions or emotionally neutral addiction indicators. Such analysis will help refine the model and improve its applicability in real-world scenarios in future research.

To address this, we applied data augmentation techniques including BERT-based generation, synonym replacement, and back translation, expanding the dataset to 6,640 comments. Nonetheless, future research should focus on collecting larger, more diverse real-world data to enhance model robustness. Although the current dataset is balanced in terms of addiction-related and awareness-oriented comments, it was compiled from publicly accessible user-generated content on digital game platforms. In future work, we aim to expand the data sources to include more linguistically varied and context-rich platforms such as forums and social media channels.

In future research, we plan to expand the dataset by incorporating content from online forums, Reddit threads, and social media platforms to enrich linguistic diversity, context coverage, and real-world representativeness.

User feedback will be collected during the pilot deployment phase to assess perceived accuracy, usefulness, and trust in the system, which will guide iterative improvements. User feedback from the deployed interface will be collected in a pilot study to evaluate user trust, perceived accuracy, and system usefulness in real-world settings.

In future iterations, the system will be validated on independent external datasets gathered from different online platforms to further assess model generalizability.

Nevertheless, ethical considerations must be prioritized. Issues surrounding user privacy, data security, and informed consent must be addressed to ensure the responsible deployment of AI in addiction detection. Implementing transparent data policies, anonymization techniques, and privacy-preserving algorithms is essential to maintaining ethical integrity. Moreover, AI-generated classifications should be interpreted within a broader diagnostic framework, integrating expert assessments and psychological evaluations to prevent misclassification and ensure responsible use.

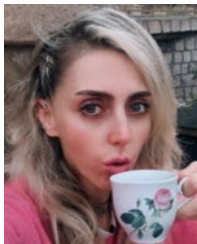
Future research should expand the dataset by incorporating in-game behavioral data, user interaction patterns, and additional linguistic features to enhance model performance and adaptability. Furthermore, continuously updating the model with real-time data could facilitate early identification of addiction symptoms and enable personalized risk profiling, providing timely interventions before addiction behaviors become severe.

REFERENCES

- [1] R. Snijders, F. Dalpiaz, S. Brinkkemper, M. Hosseini, R. Ali, and A. Ozum, "REFine: A gamified platform for participatory requirements engineering," in *Proc. IEEE 1st Int. Workshop Crowd-Based Requirements Eng. (CrowdRE)*, Aug. 2015, pp. 1–6.
- [2] M. B. Almourad, J. McAlaney, T. Skinner, M. Pleva, and R. Ali, "Defining digital addiction: Key features from the literature," *Int. J. Environ. Res. Public Health*, vol. 17, no. 12, p. 4486, Jun. 2020, doi: [10.3390/ijerph17124486](https://doi.org/10.3390/ijerph17124486).
- [3] B. Ural, E. Sezgin, and E. Yilmaz-Akyuz, "Eating behavior, sleep disorders and digital game addiction in children and determination of associated factors by CHAID analysis," *Iranian J. Public Health*, vol. 53, no. 6, pp. 1293–1303, Jun. 2024.
- [4] *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed., American Psychiatric Publishing, Washington, DC, USA, 2013, doi: [10.1176/appi.books.9780890425596](https://doi.org/10.1176/appi.books.9780890425596).
- [5] J. S. Chibbaro, L. Ricks, and B. Lanier, "Digital awareness: A model for school counselors," *J. Sch. Couns.*, vol. 17, no. 22, pp. 1–27, Jan. 2019. [Online]. Available: <https://www.jsc.montana.edu/articles/v17n22.pdf>
- [6] A. Abendroth, D. A. Parry, D. B. le Roux, and J. Gundlach, "An analysis of problematic media use and technology use addiction scales—what are they actually assessing? responsible design, implementation and use of information and communication technology," in *Proc. 19th IFIP WG 6.11 Conf. e-Business, e-Services, e-Society, Conf. Implement. Use Inf. Commun. Technol.*, Skukuza, South Africa, Cham, Switzerland: Springer, Apr. 2020, pp. 211–222, doi: [10.1007/978-3-030-45002-1_18](https://doi.org/10.1007/978-3-030-45002-1_18).
- [7] D. A. Yengin, "Digital game addiction," in *Digital Diseases*. Frankfurt am Main, Germany: Peter Lang GmbH, Internationaler Verlag der Wissenschaften, 2020, pp. 99–119. [Online]. Available: <https://www.scopus.com/inward/record.uri?eid=2-s2.0-85091748264&partnerID=40&md5=51d5eed5924673db0dea48f2e681189f>
- [8] M. Annarumma, I. Tedesco, and L. Vitale, "Mobile generation, digital devices and preschool education," *Int. J. Digit. Literacy Digit. Competence*, vol. 9, no. 4, pp. 19–32, Oct. 2018, doi: [10.4018/ijdlc.2018100102](https://doi.org/10.4018/ijdlc.2018100102).
- [9] F. D. Keya, M. M. Rahman, M. T. Nur, and M. K. Pasa, "Parenting and child's (five years to eighteen years) digital game addiction: A qualitative study in North-Western part of Bangladesh," *Comput. Hum. Behav. Rep.*, vol. 2, Aug. 2020, Art. no. 100031, doi: [10.1016/j.chbr.2020.100031](https://doi.org/10.1016/j.chbr.2020.100031).
- [10] F. Lazarinis, K. Alexandri, C. Panagiotakopoulos, and V. S. Vergyios, "Sensitizing young children on internet addiction and online safety risks through storytelling in a mobile application," *Educ. Inf. Technol.*, vol. 25, no. 1, pp. 163–174, Jan. 2020, doi: [10.1007/s10639-019-09952-w](https://doi.org/10.1007/s10639-019-09952-w).
- [11] A. Bulut and N. Tuncay, "Social media addiction and sleep problem: A structural equation modeling," in *Proc. eLearn. Softw. Educ.*, vol. 1, Apr. 2020, pp. 461–469, doi: [10.12753/2066-026x-20-060](https://doi.org/10.12753/2066-026x-20-060).
- [12] S. Gabriel, M. Nader, and M. Hütthaler, "Stop gaming! Parents' attitude towards digital game-based learning," in *Proc. 14th Int. Conf. Game Based Learn. (ECGBL)*, 2020, pp. 197–203, doi: [10.34190/GBL.20.032](https://doi.org/10.34190/GBL.20.032).
- [13] M. Carter, K. Moore, J. Mavoa, L. Gaspard, and H. Horst, "Children's perspectives and attitudes towards fortnite 'addiction,'" *Media Int. Aust.*, vol. 176, no. 1, pp. 138–151, Aug. 2020, doi: [10.1177/1329878x20921568](https://doi.org/10.1177/1329878x20921568).
- [14] N. Lee, J.-J. Kim, Y.-B. Shin, H. Eom, M.-K. Kim, S. Kyeong, Y. H. Jung, S. Min, J. H. Kwon, and E. Kim, "Choice of leisure activities by adolescents and adults with internet gaming disorder: Development and feasibility study of a virtual reality program," *JMIR Serious Games*, vol. 8, no. 4, Dec. 2020, Art. no. e18473, doi: [10.2196/18473](https://doi.org/10.2196/18473).
- [15] B. K. Erdem and R. C. Kemiksiz, "The reveal of professions being revolutionized into online careers: E-sport activities with 'generation of digital gamers'," *J. Economy Culture Soc.*, vol. 1, pp. 315–336, 2020, doi: [10.26650/JECS2019-0107](https://doi.org/10.26650/JECS2019-0107).
- [16] S. Kloker, M. L. Riegel, and C. Weinhardt, "Sensible or too sensitive? Do privacy concerns hinder the acceptance of digital solutions to treat smartphone addiction?" in *Proc. IEEE 22nd Conf. Bus. Informat. (CBI)*, vol. 2, Jun. 2020, pp. 10–19, doi: [10.1109/CBI49978.2020.10053](https://doi.org/10.1109/CBI49978.2020.10053).
- [17] A. Tutgun-Ünal, "A comparative study of social media addiction among Turkish and Korean University students," *J. Economy Culture Soc.*, vol. 62, pp. 307–322, 2020, doi: [10.26650/jecs2020-0064](https://doi.org/10.26650/jecs2020-0064).
- [18] A. Malik, A. P. Nanda, and R. Kumra, "Children in the digital world: Exploring the role of parental-child attachment features in excessive online gaming," *Young Consumers*, vol. 21, no. 3, pp. 335–350, Jun. 2020, doi: [10.1108/yc-01-2020-1090](https://doi.org/10.1108/yc-01-2020-1090).
- [19] I. Pogozhina, A. Podolsky, O. Idobaeva, and T. Podolskaya, "Behavioral and motivational patterns of internet users: A logico-categorical analysis," *Voprosy Obrazovaniya/Educ. Stud. Moscow*, vol. 3, pp. 60–94, Sep. 2020, doi: [10.17323/1814-9545-2020-3-60-94](https://doi.org/10.17323/1814-9545-2020-3-60-94).
- [20] T. Schatto-Eckrodt, R. Janzik, F. Reer, S. Boberg, and T. Quandt, "A computational approach to analyzing the Twitter debate on gaming disorder," *Media Commun.*, vol. 8, no. 3, pp. 205–218, Aug. 2020, doi: [10.17645/mac.v8i3.3128](https://doi.org/10.17645/mac.v8i3.3128).
- [21] G. F. Romashkina and R. R. Khuziakmetov, "The risks of internet addiction: Structure and characteristics of perception," *Obrazovanie I Nauka*, vol. 22, no. 8, pp. 108–134, 2020, doi: [10.17853/1994-5639-2021-8-108-134](https://doi.org/10.17853/1994-5639-2021-8-108-134).
- [22] F. Khalid and T. El-Maliki, "Teachers' experiences in the development of digital storytelling for cyber risk awareness," *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, no. 2, pp. 186–191, 2020.
- [23] A. Aborujilah, R. Mohammad, A. Z. Al-Othmani, M. N. Husen, Z. A. Long, N. A. Ali, and V. A. Murugan, "Adoption and continuance intention model of applying telemedicine technology in digital games addiction," in *Proc. 14th Int. Conf. Ubiquitous Inf. Manage. Commun. (IMCOM)*, Jan. 2020, pp. 1–9, doi: [10.1109/IMCOM48794.2020.9001732](https://doi.org/10.1109/IMCOM48794.2020.9001732).
- [24] T.-C. Yang, M. C. Chen, and S. Y. Chen, "Incorporating digital games into anti-drug material: Non-drug-addicted learners vs. drug-addicted learners," *Universal Access Inf. Soc.*, vol. 19, no. 1, pp. 29–40, Mar. 2020, doi: [10.1007/s10209-018-0624-6](https://doi.org/10.1007/s10209-018-0624-6).
- [25] X. Hu and H. Zhang, "Transforming from addicted video gamer to doctoral candidate: An autoethnographic reflection," *Qualitative Rep.*, vol. 27, no. 11, pp. 2404–2418, 2022, doi: [10.46743/2160-3715/2022.5528](https://doi.org/10.46743/2160-3715/2022.5528).
- [26] W. L. Lee, P. N. I. M. S. Rambiar, N. Q. B. Rosli, M. S. Nurmal, S. S. S. Abdullah, and M. Danaee, "Impact of increased digital use and internet gaming on nursing students' empathy: A cross-sectional study," *Nurse Educ. Today*, vol. 119, Dec. 2022, Art. no. 105563, doi: [10.1016/j.nedt.2022.105563](https://doi.org/10.1016/j.nedt.2022.105563).
- [27] I. Cheah, A. S. Shimul, and I. Phau, "Motivations of playing digital games: A review and research agenda," *Psychol. Marketing*, vol. 39, no. 5, pp. 937–950, May 2022, doi: [10.1002/mar.21631](https://doi.org/10.1002/mar.21631).
- [28] S. Chatterjee, R. Chaudhuri, and D. Vrontis, "Examining the antecedents and consequences of addiction to mobile games: An empirical study," *Inf. Syst. e-Bus. Manage.*, vol. 23, no. 1, pp. 241–260, Mar. 2025, doi: [10.1007/s10257-022-00614-y](https://doi.org/10.1007/s10257-022-00614-y).
- [29] W. H. A. H. Elkhateeb, Z. M. A. Azim, and M. H. R. Khalaf, "Electronic games and their relationship to social responsibility and information processing among university students," *Int. J. Comput. Sci. Netw. Secur. (IJCSNS)*, vol. 22, no. 3, pp. 413–416, 2022, doi: [10.22937/IJCSNS.2022.22.3.52](https://doi.org/10.22937/IJCSNS.2022.22.3.52).
- [30] A. K. Alan, E. T. Kabadayi, and N. C. Aksoy, "Replaying online games for flow experience and outcome expectations: An exploratory study for the moderating role of external locus of control based on Turkish gamers' evaluations," *Entertainment Comput.*, vol. 40, Jan. 2022, Art. no. 100460, doi: [10.1016/j.entcom.2021.100460](https://doi.org/10.1016/j.entcom.2021.100460).

- [31] P. Kaimara, A. Oikonomou, and I. Deliyannis, "Could virtual reality applications pose real risks to children and adolescents? A systematic review of ethical issues and concerns," *Virtual Reality*, vol. 26, no. 2, pp. 697–735, Jun. 2022, doi: [10.1007/s10055-021-00563-w](https://doi.org/10.1007/s10055-021-00563-w).
- [32] Ç. Geniş and S. Ayaz-Alkaya, "Digital game addiction, social anxiety, and parental attitudes in adolescents: A cross-sectional study," *Children Youth Services Rev.*, vol. 149, Jun. 2023, Art. no. 106931, doi: [10.1016/j.chilcyouth.2023.106931](https://doi.org/10.1016/j.chilcyouth.2023.106931).
- [33] A. Alrobai, K. Phalp, and R. Ali, "Digital addiction: A requirements engineering perspective," in *Proc. Int. Working Conf. Requirements Eng., Found. Softw. Quality*, vol. 8396, 2014, pp. 112–118, doi: [10.1007/978-3-319-05843-6_9](https://doi.org/10.1007/978-3-319-05843-6_9).
- [34] R. Ali, N. Jiang, K. Phalp, S. Muir, and J. McAlaney, "The emerging requirement for digital addiction labels," in *Proc. Int. Working Conf. Requirements Eng., Found. Softw. Quality*, vol. 9013, Jan. 2015, pp. 198–213, doi: [10.1007/978-3-319-16101-3_13](https://doi.org/10.1007/978-3-319-16101-3_13).
- [35] A. Alrobai, J. McAlaney, K. Phalp, and R. Ali, "Online peer groups as a persuasive tool to combat digital addiction," *Persuasive Technol.*, vol. 9638, pp. 288–300, Jan. 2016, doi: [10.1007/978-3-319-31510-2_25](https://doi.org/10.1007/978-3-319-31510-2_25).
- [36] R. Ali, "Digital motivation, digital addiction and responsibility requirements," in *Proc. 1st Int. Workshop Affect. Comput. Requirements Eng. (AffectRE)*, Aug. 2018, p. 27, doi: [10.1109/AffectRE.2018.00011](https://doi.org/10.1109/AffectRE.2018.00011).
- [37] Z. Hazar, "Children's digital game addiction and opinions about their parents' playing digital games (a mixed method study)," *J. Educ. Training Stud.*, vol. 7, no. 1, p. 85, Dec. 2018, doi: [10.11114/jets.v7i1.3785](https://doi.org/10.11114/jets.v7i1.3785).
- [38] M. Aldhayan, S. Cham, T. Kostoulas, M. B. Al-Mourad, and R. Ali, "Online peer support groups to combat digital addiction: User acceptance and rejection factors," *Adv. Intell. Syst. Comput.*, vol. 932, pp. 139–150, Jan. 2019.
- [39] J. Ruedas, S. Serrate, and J. M. Muñoz, "Hyper-connected youth: The standpoint of families facing their children's virtual interaction," in *Proc. 8th Int. Conf. Technolog. Ecosyst. Enhancing Multiculturality*, Oct. 2020, pp. 935–939, doi: [10.1145/3434780.3436628](https://doi.org/10.1145/3434780.3436628).
- [40] S. Bhagat, E. J. Jeong, and D. J. Kim, "The role of individuals' need for online social interactions and interpersonal incompetence in digital game addiction," *Int. J. Hum.-Comput. Interact.*, vol. 36, no. 5, pp. 449–463, Mar. 2020, doi: [10.1080/10447318.2019.1654696](https://doi.org/10.1080/10447318.2019.1654696).
- [41] A. Wijnarko and A. D. Hadisumarto, "Online video games as distribution channel for retail brand voucher," in *ACM Int. Conf. Proc. Ser.*, 2020, pp. 7–14, doi: [10.1145/3409929.3409931](https://doi.org/10.1145/3409929.3409931).
- [42] H. Li, X. Zhang, Y. Cao, and G. Zhang, "Potential protection of computer gaming against mental health issues: Evidence from a Mendelian randomization study," *Comput. Hum. Behav.*, vol. 144, Jul. 2023, Art. no. 107722, doi: [10.1016/j.chb.2023.107722](https://doi.org/10.1016/j.chb.2023.107722).
- [43] K. Taylor, "The social diagnoses of digital addictions: Technophobic ambivalences, the limits of the natural and imperatives of self-governance in the information age," *Sociol. Health Illness*, vol. 46, no. S1, pp. 18–36, Mar. 2024, doi: [10.1111/1467-9566.13624](https://doi.org/10.1111/1467-9566.13624).
- [44] A. Gul and H. Gul, "Sluggish cognitive tempo (cognitive disengagement syndrome) symptoms are more associated with a higher risk of internet addiction and internet gaming disorder than ADHD symptoms: A study with medical students and resident doctors," *Res. Develop. Disabilities*, vol. 139, Aug. 2023, Art. no. 104557, doi: [10.1016/j.ridd.2023.104557](https://doi.org/10.1016/j.ridd.2023.104557).
- [45] C. Chen, "Playing mobile games for stress recovery purposes: A survey of Chinese adolescents," *Telematics Informat.*, vol. 56, Jan. 2021, Art. no. 101481, doi: [10.1016/j.tele.2020.101481](https://doi.org/10.1016/j.tele.2020.101481).
- [46] J. Heo and E. Park, "Are you satisfied or satiated by the games you play? An empirical study on the relationship between play patterns and game purchase behavior," *J. Retailing Consum. Services*, vol. 61, Jan. 2021, Art. no. 102551, doi: [10.1016/j.jretconser.2021.102551](https://doi.org/10.1016/j.jretconser.2021.102551).
- [47] A. D. Keskin and N. Aral, "Positive and negative aspects of digital games from the perspective of adolescents," *Turk. Psychol. Couns. Guid. J.*, vol. 12, no. 65, pp. 543–557, 2022, doi: [10.17066/TPDRD.12241172](https://doi.org/10.17066/TPDRD.12241172).
- [48] N. Kazaz, T. Dilci, and T. Karadağ, "Effects of digital media on education (meta-thematic analysis)," *Int. J. Emerg. Technol. Learn. (iJET)*, vol. 17, no. 16, pp. 222–242, Aug. 2022, doi: [10.3991/iJET.v17i16.32181](https://doi.org/10.3991/iJET.v17i16.32181).
- [49] V. Bezençon, A. De Santo, A. Holzer, and B. Lanz, "Escape addict: A digital escape room for the prevention of addictions and risky behaviors in schools," *Comput. Educ.*, vol. 200, Jul. 2023, Art. no. 104798, doi: [10.1016/j.compedu.2023.104798](https://doi.org/10.1016/j.compedu.2023.104798).
- [50] B. I. Davidson, H. Shaw, and D. A. Ellis, "Fuzzy constructs in technology usage scales," *Comput. Hum. Behav.*, vol. 133, Aug. 2022, Art. no. 107206, doi: [10.1016/j.chb.2022.107206](https://doi.org/10.1016/j.chb.2022.107206).
- [51] D. M. Miranda, R. M. Pontes, and T. de Gois Ribeiro Darin, "It's dark but just a game: Towards an ethical and healthy game design practice," in *Proc. 21st Brazilian Symp. Hum. Factors Comput. Syst.*, Oct. 2022, pp. 1–7, doi: [10.1145/3554364.3559144](https://doi.org/10.1145/3554364.3559144).
- [52] K. Lindenberg and U. Basten, "Development of intelligence in the context of digital media use," in *Intelligence—Theories and Applications*, R. M. Holm-Hadulla, J. Funke, and M. Wink, Eds., Cham, Switzerland: Springer, 2022, pp. 161–184, doi: [10.1007/978-3-031-04198-3_9](https://doi.org/10.1007/978-3-031-04198-3_9).
- [53] A. Kaya and F. Pazarcıkcı, "Structural equation modeling analysis of risk factors for digital game addiction in adolescents: A web-based study," *Arch. Psychiatric Nursing*, vol. 43, pp. 22–28, Apr. 2023, doi: [10.1016/j.apnu.2022.12.031](https://doi.org/10.1016/j.apnu.2022.12.031).
- [54] M. E. Green and M. N. McNeese, "Factors that predict digital game play," *Howard J. Commun.*, vol. 19, no. 3, pp. 258–272, Jul. 2008, doi: [10.1080/10646170802218321](https://doi.org/10.1080/10646170802218321).
- [55] G. Porter, V. Starcevic, D. Berle, and P. Fenech, "Recognizing problem video game use," *Austral. New Zealand J. Psychiatry*, vol. 44, no. 2, pp. 120–128, Feb. 2010, doi: [10.3109/00048670903279812](https://doi.org/10.3109/00048670903279812).
- [56] D. A. Gentile, H. Choo, A. Liau, T. Sim, D. Li, D. Fung, and A. Khoo, "Pathological video game use among youths: A two-year longitudinal study," *Pediatrics*, vol. 127, no. 2, pp. e319–e329, Feb. 2011, doi: [10.1542/peds.2010-1353](https://doi.org/10.1542/peds.2010-1353).
- [57] S. Tokar and M. H. Baturay, "Antecedents and consequences of game addiction," *Comput. Hum. Behav.*, vol. 55, pp. 668–679, Feb. 2016, doi: [10.1016/j.chb.2015.10.002](https://doi.org/10.1016/j.chb.2015.10.002).
- [58] E. F. Yang, R. Kornfield, Y. Liu, M.-Y. Chih, P. Sarma, D. Gustafson, J. Curtin, and D. Shah, "Using machine learning of online expression to explain recovery trajectories: Content analytic approach to studying a substance use disorder forum," *J. Med. Internet Res.*, vol. 25, Aug. 2023, Art. no. e45589, doi: [10.2196/45589](https://doi.org/10.2196/45589).
- [59] V. Stavropoulos, M. D. Griffiths, T. L. Burleigh, D. J. Kuss, Y. Y. Doh, and R. Gomez, "Flow on the internet: A longitudinal study of internet addiction symptoms during adolescence," *Behav. Inf. Technol.*, vol. 37, no. 2, pp. 159–172, Feb. 2018, doi: [10.1080/0144929x.2018.1424937](https://doi.org/10.1080/0144929x.2018.1424937).
- [60] P. C. Morgado, M. Carusso, L. A. Alemany, and L. Acion, "Practical foundations of machine learning for addiction research. Part I. Methods and techniques," *Amer. J. Drug Alcohol Abuse*, vol. 48, no. 3, pp. 260–271, May 2022, doi: [10.1080/00952990.2021.1995739](https://doi.org/10.1080/00952990.2021.1995739).
- [61] A. Bouhadja and A. Bouramoul, "A review on recent machine learning applications for addiction disorders," in *Proc. 4th Int. Conf. Pattern Anal. Intell. Syst. (PAIS)*, Oct. 2022, pp. 1–8, doi: [10.1109/PAIS56586.2022.9946888](https://doi.org/10.1109/PAIS56586.2022.9946888).
- [62] C. S. Wilkinson, M. Á. Luján, C. Hales, K. M. Costa, V. G. Fiore, L. A. Knackstedt, and H. Kober, "Listening to the data: Computational approaches to addiction and learning," *J. Neurosci.*, vol. 43, no. 45, pp. 7547–7553, Nov. 2023, doi: [10.1523/jneurosci.1415-23.2023](https://doi.org/10.1523/jneurosci.1415-23.2023).
- [63] W. Abada and A. Bouramoul, "Using machine learning techniques to predict people at-risk for drug addiction: A Bayesian-based model," in *Proc. 4th Int. Conf. Pattern Anal. Intell. Syst. (PAIS)*, Constantine, Algeria, Oct. 2022, pp. 1–7, doi: [10.1109/pais56586.2022.9946914](https://doi.org/10.1109/pais56586.2022.9946914).
- [64] S. Sujana, A. N. Kumar, L. A. S. Charan, G. Nishitha, and M. Ratnababu, "Comparative analysis of deep learning techniques for drug addiction," in *Proc. IEEE 9th Int. Conf. Conver. Technol. (I2CT)*, Pune, India, 2024, pp. 1–6, doi: [10.1109/I2CT61223.2024.10544100](https://doi.org/10.1109/I2CT61223.2024.10544100).
- [65] U. G. H. T. Kumara, S. S. A. Siriwardana, L. Weerasinghe, R. A. K. I. Shavindi, H. P. R. C. Chiranjeeva, and S. Siriwardana, "A machine learning approach to analyze the drug addiction," in *Proc. 5th Int. Conf. Advancements Comput. (ICAC)*, Dec. 2023, pp. 113–118, doi: [10.1109/ICAC60630.2023.10417256](https://doi.org/10.1109/ICAC60630.2023.10417256).
- [66] A. Singh and S. Babbar, "Detecting internet addiction disorder using Bayesian networks," in *Data Science and Analytics*, B. Panda, S. Sharma, and N. R. Roy, Eds., Singapore: Springer, 2018, pp. 80–95, doi: [10.1007/978-981-10-8527-7_8](https://doi.org/10.1007/978-981-10-8527-7_8).
- [67] Y. Du, "Research on lexical data set analysis based on decision tree model," *Highlights Sci. Eng. Technol., AIDML*, vol. 57, pp. 108–114, 2023, doi: [10.54097/hset.v57i.9987](https://doi.org/10.54097/hset.v57i.9987).
- [68] A. N. Tusher, S. Islam, M. T. Islam, M. S. R. Sammy, M. S. Rahman, and M. S. Sadik, "User perspective Bangla sentiment analysis for online gaming addiction using machine learning," in *Proc. 6th Int. Conf. I-SMAC (IoT Social, Mobile, Analytics Cloud) (I-SMAC)*, Nov. 2022, pp. 538–543, doi: [10.1109/I-SMAC55078.2022.9987343](https://doi.org/10.1109/I-SMAC55078.2022.9987343).

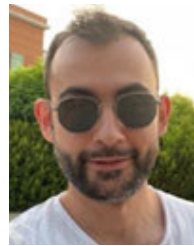
- [69] D. Kornev, R. Sadeghian, S. Nwoji, Q. He, A. Gandjbakhche, and S. Aram, "Machine learning-based gaming behavior prediction platform," *Neuroergonomics Cognit. Eng.*, vol. 42, pp. 122–128, Jan. 2022, doi: [10.54941/ahfe1001826](https://doi.org/10.54941/ahfe1001826).
- [70] M. Güllü, F. H. Yagin, I. Gocer, H. Yapici, E. Ayyildiz, F. M. Clemente, L. P. Ardigo, A. K. Zadeh, P. Prieto-González, and H. Nobari, "Exploring obesity, physical activity, and digital game addiction levels among adolescents: A study on machine learning-based prediction of digital game addiction," *Frontiers Psychol.*, vol. 14, Mar. 2023, Art. no. 1097145, doi: [10.3389/fpsyg.2023.1097145](https://doi.org/10.3389/fpsyg.2023.1097145).
- [71] V. Kuna, "Machine learning models in predicting gaming popularity: A comparative analysis," *Int. J. Multidisciplinary Res.*, vol. 6, no. 3, pp. 1–7, Jun. 2024, doi: [10.36948/ijfmr.2024.v06i03.14362](https://doi.org/10.36948/ijfmr.2024.v06i03.14362).
- [72] R. Narula, R. Sharma, and A. Kalra, "Accuracy comparison of data mining algorithms for internet gaming disorder classification," *J. Addict. Behav. INJURY*, vol. 2, no. 4, pp. 455–467, 2023, doi: [10.58631/injury.v2i4.62](https://doi.org/10.58631/injury.v2i4.62).
- [73] J. Choi and D. Seo, "View of internet gambling through text mining analysis," *Proc. Nat. Res.*, vol. 13, no. S03, p. 93, 2022, doi: [10.47750/pnr.2022.13.S03.093](https://doi.org/10.47750/pnr.2022.13.S03.093).
- [74] E. K. Ozyirmidokuz and B. A. Celik, "Expert-reviewed digital game addiction dataset (Balanced & Validated)," *Mendeley Data*, vol. 1, 2024, doi: [10.17632/p4fhnnbp6x.1](https://doi.org/10.17632/p4fhnnbp6x.1).
- [75] A. Parashar, A. Parashar, W. Ding, M. Shabaz, and I. Rida, "Data preprocessing and feature selection techniques in gait recognition: A comparative study of machine learning and deep learning approaches," *Pattern Recognit. Lett.*, vol. 172, pp. 65–73, Aug. 2023, doi: [10.1016/j.patrec.2023.05.021](https://doi.org/10.1016/j.patrec.2023.05.021).
- [76] E. K. Ozyirmidokuz and B. A. Celik, "Machine learning models application through text dataset-2," *Zenodo*, 2024, doi: [10.5281/zenodo.13227780](https://doi.org/10.5281/zenodo.13227780).



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